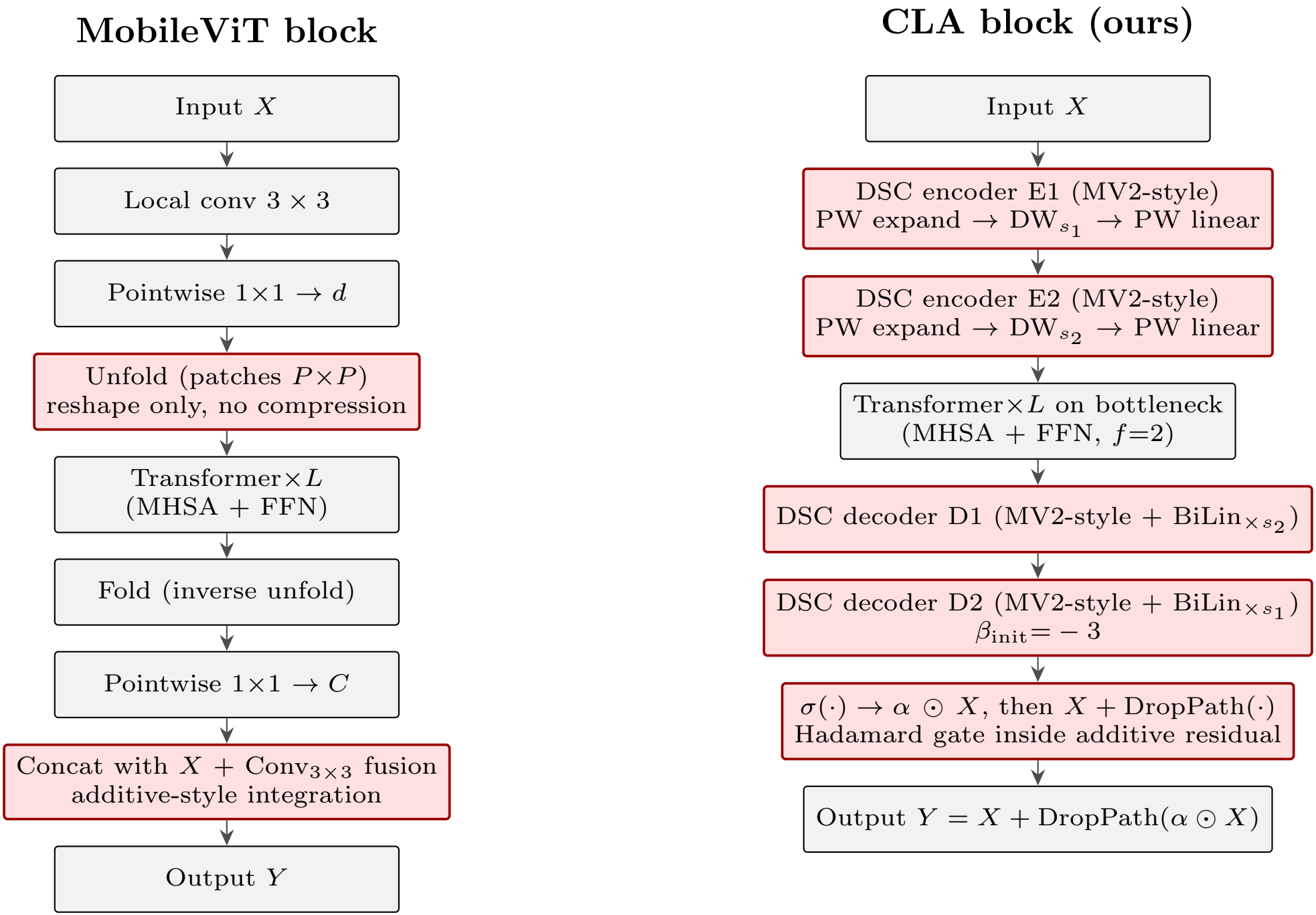

MobileViT block
Input X
Local conv 3×3
Pointwise $1\times1 \rightarrow d$
Unfold (patches $P\times P$)
reshape only, no compression
Transformer$\times L$
(MHSA + FFN)
Fold (inverse unfold)
Pointwise $1\times1 \rightarrow C$
Concat with X + $\mathrm{Conv}_{3\times3}$ fusion
additive-style integration
Output Y
CLA block (ours)
Input X
DSC encoder E1 (MV2-style)
PW expand $\rightarrow \mathrm{DW}_{s_1} \rightarrow$ PW linear
DSC encoder E2 (MV2-style)
PW expand $\rightarrow \mathrm{DW}_{s_2} \rightarrow$ PW linear
Transformer$\times L$ on bottleneck
(MHSA + FFN, f=2)
DSC decoder D1 (MV2-style + $\mathrm{BiLin}_{\times s_2}$)
DSC decoder D2 (MV2-style + $\mathrm{BiLin}_{\times s_1}$)
$\beta_{\mathrm{init}} = -3$
$\sigma(\cdot) \rightarrow \alpha \odot X$, then $X + \mathrm{DropPath}(\cdot)$
Hadamard gate inside additive residual
Output $Y = X + \mathrm{DropPath}(\alpha \odot X)$